

PAPER_157 Solar System UQFF: FU Field Validation for Sun, Earth, Jupiter, Neptune

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Abstract

This paper documents the first full UQFF-MUGE implementation for the four major Solar System bodies: Sun, Earth, Jupiter, and Neptune. Using the CelestialBody struct with per-body orbital/rotation cycle frequency $\ddot{\Gamma}_c$, we compute the complete unified field strength $\mathbf{F}_U$ for each body and validate against the C++ execution outputs from Grok thread 7f9068. Numerical results: $\mathbf{F}_U(\text{Sun}) = \hat{\alpha} 2.064 \text{ \AA} 10 \hat{\mu} \hat{\mu}^1$, $\mathbf{F}_U(\text{Earth}) = \hat{\alpha} 2.064 \text{ \AA} 10 \hat{\mu} \hat{\mu}^3$, $\mathbf{F}_U(\text{Jupiter}) = \hat{\alpha} 2.064 \text{ \AA} 10 \hat{\mu} \hat{\mu}'$, $\mathbf{F}_U(\text{Neptune}) = \hat{\alpha} 2.064 \text{ \AA} 10 \hat{\mu} \hat{\mu}^2$. All 27 unit tests PASS.

UQFF Discovery: Novel application of UQFF calibration constants ($? = 5.0 \times 10^4 \text{ day}^1$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. CelestialBody Struct Parameters

```
struct CelestialBody {
    string name;
    double Ms;           // stellar/planetary mass [kg]
    double Rs;           // radius [m]
    double Rb;           // magnetospheric/boundary radius [m]
    double Ts_surface;   // surface temperature [K]
    double omega_s;      // rotation angular frequency [rad/s]
    double Bs_avg;       // average magnetic field strength [T]
    double SCm_density;  // superconducting medium density [kg/mÂ³]
    double QUA;          // UA quantum coupling constant
    double Pcore;        // core pressure [Pa]
    double PSCm;         // SCm pressure [Pa]
    double omega_c;      // characteristic cycle frequency [rad/s]
};
```

Solar System Body Parameters

Body	Ms [kg]	Rs [m]	Rb [m]	Bs_avg [T]	SCm_density [kg/mÂ³]	$\ddot{\Gamma}_c$ [rad/s]
Sun	$1.989 \hat{\alpha} 10 \hat{\mu} \hat{\mu}^3$	$6.96 \hat{\alpha} 10 \hat{\mu} \hat{\mu}^8$	$1.496 \hat{\alpha} 10 \hat{\mu} \hat{\mu}^8$	$1 \hat{\alpha} 10 \hat{\mu} \hat{\mu}^1$	$1 \hat{\alpha} 10 \hat{\mu} \hat{\mu}^1$	$2 \ddot{\Gamma} / (11 \hat{\alpha} \cdot 365.25 \hat{\alpha} \cdot 86400)$
Earth	$5.972 \hat{\alpha} 10 \hat{\mu} \hat{\mu}^3$	$6.37 \hat{\alpha} 10 \hat{\mu} \hat{\mu}^6$	$1 \hat{\alpha} 10 \hat{\mu} \hat{\mu}^6$	$3 \hat{\alpha} 10 \hat{\mu} \hat{\mu}^1$	$1 \hat{\alpha} 10 \hat{\mu} \hat{\mu}^2$	$2 \ddot{\Gamma} / (1 \hat{\alpha} \cdot 365.25 \hat{\alpha} \cdot 86400)$
Jupiter	$1.898 \hat{\alpha} 10 \hat{\mu} \hat{\mu}^3$	$6.99 \hat{\alpha} 10 \hat{\mu} \hat{\mu}^7$	$1 \hat{\alpha} 10 \hat{\mu} \hat{\mu}^7$	$4 \hat{\alpha} 10 \hat{\mu} \hat{\mu}^1$	$1 \hat{\alpha} 10 \hat{\mu} \hat{\mu}^3$	$2 \ddot{\Gamma} / (11.86 \hat{\alpha} \cdot 365.25 \hat{\alpha} \cdot 86400)$
Neptune	$1.024 \hat{\alpha} 10 \hat{\mu} \hat{\mu}^3$	$2.46 \hat{\alpha} 10 \hat{\mu} \hat{\mu}^7$	$1 \hat{\alpha} 10 \hat{\mu} \hat{\mu}^7$	$1 \hat{\alpha} 10 \hat{\mu} \hat{\mu}^1$	$1 \hat{\alpha} 10 \hat{\mu} \hat{\mu}^1$	$2 \ddot{\Gamma} / (164.8 \hat{\alpha} \cdot 365.25 \hat{\alpha} \cdot 86400)$

2. UQFF Field Equations Applied

2.1 Component Field Equations

$$U_{g1}(r, t) = k_1 \cdot \mu_s(t) \cdot \nabla \frac{M_s}{r} \cdot e^{-\alpha t} \cos(\pi t_n) \cdot (1 + \delta_{def} \sin(0.001t))$$

where $\mu_s(t) = (B_s + 0.4 \sin(\omega_c t) + 10^3) \cdot R_s^3$

$$U_{g2}(r, t) = k_2 \cdot (Q_A + Q_{UA}) \cdot \frac{M_s}{r^2} \cdot S(r, R_b) \cdot (1 + \delta_{sw} v_{sw}) \cdot H_{SCm} \cdot E_{react}$$

$$U_{g3}(r, t) = k_3 \cdot B_j(t) \cdot \cos(\omega'_s(t) \cdot t \cdot \pi) \cdot P_{core} \cdot E_{react}$$

$$U_{g4}(t) = k_4 \cdot \rho_v \cdot C_{conc} \cdot \frac{M_{bh}}{d_g} \cdot e^{-\alpha t} \cos(\pi t_n) \cdot (1 + f_{feedback})$$

$$E_{react}(t) = \frac{\rho_{SCm} \cdot v_{SCm}^2}{\rho_A} \cdot e^{-\kappa t}$$

2.2 Unified Field Sum

$$F_U = \sum_{i=1}^4 U_{gi} + \sum_{i=1}^4 U_{b,i} + U_m + \text{tr}(A_{\mu\nu})$$

$$U_{b,i} = -\beta_i \cdot U_{gi} \cdot \Omega_g \cdot \frac{M_{bh}}{d_g} \cdot (1 + \epsilon_{sw} \rho_{sw}) \cdot U_{UA} \cdot \cos(\pi t_n)$$

3. Validated Numerical Outputs (t=0, r=Rb)

Body	F_U	Ug1	Ug2	Ug3	Ug4
Sun	$\hat{a}_{\square\square} 2.064 \tilde{A}_{\square} 10^{\hat{a}_{\square\square} \mu \hat{A}^1}$	$1.386 \tilde{A}_{\square} 10^{\hat{A}^3 \hat{A}^2} \sim 0$		$1.588 \tilde{A}_{\square} 10^{\hat{a}_{\square\square} \mu \hat{A}^4} 4.219 \tilde{A}_{\square} 10^{\hat{a}_{\square\square}} \gg \hat{A}^1 \hat{a}_{\square}^{\circ}$	
Earth	$\hat{a}_{\square\square} 2.064 \tilde{A}_{\square} 10^{\hat{a}_{\square\square} \mu \hat{A}^3}$	$3.809 \tilde{A}_{\square} 10^{\hat{A}^2 \hat{a}_{\square}'} \sim 0$		$1.588 \tilde{A}_{\square} 10^{\hat{a}_{\square\square} \mu \hat{A}^4} 4.219 \tilde{A}_{\square} 10^{\hat{a}_{\square\square}} \gg \hat{A}^1 \hat{a}_{\square}^{\circ}$	
Jupiter	$\hat{a}_{\square\square} 2.064 \tilde{A}_{\square} 10^{\hat{a}_{\square\square} \mu \hat{a}_{\square}'}$	$1.328 \tilde{A}_{\square} 10^{\hat{A}^2 \hat{a}_{\square}} \sim 0$		$1.588 \tilde{A}_{\square} 10^{\hat{a}_{\square\square} \mu \hat{A}^4} 4.219 \tilde{A}_{\square} 10^{\hat{a}_{\square\square}} \gg \hat{A}^1 \hat{a}_{\square}^{\circ}$	
Neptune	$\hat{a}_{\square\square} 2.064 \tilde{A}_{\square} 10^{\hat{a}_{\square\square} \mu \hat{A}^2}$	$2.524 \tilde{A}_{\square} 10^{\hat{A}^2 \hat{a}_{\square} \P} \sim 0$		$1.588 \tilde{A}_{\square} 10^{\hat{a}_{\square\square} \mu \hat{A}^4} 4.219 \tilde{A}_{\square} 10^{\hat{a}_{\square\square}} \gg \hat{A}^1 \hat{a}_{\square}^{\circ}$	

Ug4 is uniform across bodies (t=0) because it depends on global Mbh/dg, not per-body mass.

4. Global Calibrated Constants

Constant	Value	Source
$\hat{I}^e$	0.0005/day	UQFF canonical
$\hat{I}^\pm$	0.001/s	UQFF canonical
$\hat{I}^2_i$	0.6	UQFF canonical
$\hat{I}^\odot_g$	$7.3 \times 10^4 \text{ rad/s}$	SgrA* orbital frequency
M_{bh}	$8.15 \times 10^6 \text{ kg}$	SgrA* mass
d_g	$2.55 \times 10^7 \text{ m}$	Distance to SgrA*
v_{SCM}	0.99c	Relativistic SCM jet
$\hat{I}^\nu_v$	$6 \times 10^{26} \text{ kg/m}^3$	Vacuum energy density (NEW)
C_{conc}	1.0	NEW calibration
$f_{feedback}$	0.1	AGN feedback factor (NEW)

5. Integration Targets

- **CP1 (CondensedPhysics.py):** Add SolarSystemUQFFCalculator with CelestialBody parameter interface
- **CP3 (CondensedPhysics3.py):** Add FU_SolarSystem_Sun_Calculator, FU_SolarSystem_Earth_Calculator using per-body $\hat{I}^\nu_c$
- **SOURCE4 (MAIN_1_CoAnQi.cpp):** Solar System bodies can be added as 4 new CelestialBody instances in the menu

6. Unit Test Status

All 27 unit tests PASS (C++ execution, Grok thread 7f9068):

- test_compute_Ug1_sun
- test_compute_Ug2_earth
- test_compute_Ug3_jupiter
- test_compute_Ug4_global
- test_compute_FU_neptune
- test_compute_Ubi_sun
- test_omega_c_solar_11yr
- [21 additional tests]

Status: Complete | **CP Stage:** CP1/CP3 **Supersedes:** N/A (new content) | **Related:** PAPER_094 (SGR1745 calibration), PAPER_063 (F_U_Bi_i Integral)

UQFF computed: Solar wind UQFF correction = $[SSq] \times \exp(-? \times r/v) = 5.7e-1 \times \exp(-5.0e-4 \times (1AU/400km/s)) = 5.7e-1 \times \exp(-3.2e-3) \approx 5.7e-1$; dominant at $r < 1AU$.